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ORIGINAL ARTICLE

Fragility of Indonesian houses: scenario damage analysis of the 2006 Yogyakarta and 2009 Padang earthquakes

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Abstract
Indonesia is located in one of the most seismically active regions in the world and often experiences damaging earthquakes. In the past the housing sector has sustained higher earthquake related damage and losses than other sectors. This is often attributed to the fact that the most common houses in Indonesia are non-engineered, built with poor quality workmanship, poor quality materials and without resilient seismic design features. However little effort has been made to quantify how fragile Indonesian houses are, or how their fragility may vary according to the population density or relative wealth of a region. It is not possible to derive empirical fragility functions for Indonesia due to insufficient damage data. The aim of this study is to determine whether existing earthquake fragility functions can be applied to common house types in Indonesia. Scenario damage analyses simulating the 2006 Yogyakarta and 2009 Padang events were undertaken several times testing different fragility functions. The simulated damage results were then compared to the damage observed post event to determine whether an accurate damage prediction could be achieved. It was found that the common house types in Yogyakarta and Central Java vary according to age of construction, location and relative wealth of a region and can be reasonably well represented by existing fragility functions. However, the houses in Padang and surrounding West Sumatra did not vary in a predictable manner and are more fragile than anticipated. Therefore, the fragility of the most common house types in Indonesia differs between Central Java and West Sumatra. This has important implications for seismic damage and risk assessment undertaken in Indonesia.

Keywords Fragility · Exposure · Damage assessment · Seismic risk · Indonesia · Residential buildings · Modeling · Scenario analysis

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1 Introduction

Indonesia is located in one of the most seismically active regions in the world and has experienced many damaging and fatal earthquakes in the past (Irsyam et al. 2020). The housing sector is particularly vulnerable to earthquakes, due to the prevalence of non-engineered houses which are not required to conform to the building code (Idham 2019). These houses account for between 70 and 90% of residential buildings and are often constructed with poor quality workmanship, poor quality materials and without resilient seismic design provisions (Boen and Pribadi 2007; Idham 2019). In the 2006 Yogyakarta and 2009 Padang earthquakes it was noted that the level of damage sustained was disproportionate to the magnitude of the earthquake, due to poor quality construction in the housing sector (BAPPENAS 2006; BNBP 2009). Despite the known hazard and vulnerable building stock there have been limited attempts to quantify the fragility of houses in Indonesia or determine whether the fragility varies across the country.

Fragility functions are a key component of damage and risk analysis, mathematically defining the relationship between the ground motion intensity and probability of damage occurring. Functions are generally unique to a building type and can be derived empirically, analytically, by expert opinion or a combination of methods (Kappos et al. 2006). All derivation methods require significant quantities of data including ground motion records; post disaster damage to buildings; and, the types of buildings defined by construction methods, materials, ages and building code compliance (Hoyos and Hernández 2021; Tocchi et al. 2022). In Indonesia, and other locations where such data are limited or nonexistent, it is difficult to develop localized functions and thus functions developed for other locations are commonly used (Hoyos and Hernández 2021). This practice introduces an unknown degree of uncertainty into any damage or risk assessment (Crowley et al. 2008; Selva et al. 2013). Understanding and accounting for this uncertainty is a current challenge in earthquake damage and risk assessments (Silva et al. 2019).

In this study we use scenario damage analysis to determine if existing fragility functions can be used to represent the housing stock in Indonesia. We consider the only two historical earthquakes which have significant post disaster damage data for verification of simulation results, the 2006 Yogyakarta and 2009 Padang earthquakes on the islands of Java and Sumatra, respectively. A suite of existing fragility functions is tested to determine which perform the best in replicating the observed damage. A secondary aim is to assess the sensitivity of scenario damage simulations to input model selection for fragility functions, ground motion models and the Vs30 models. The results are important for Indonesia other locations with limited data availability to understand how to accurately represent the fragility of housing stock in Indonesia and the impact model selection decisions can have on the simulated damage. The first part of the paper addresses the selection of ground motion model, Vs30 and the development of the residential exposure models and the uncertainty associated with these. The second part of the paper focuses on the sensitivity of damage results to input model selection and assesses both the performance of the available fragility functions, and whether there is significant variability of fragility between Java and Sumatra.

2 Case studies

The 2006 Yogyakarta and 2009 Padang earthquakes have reasonably detailed damage data that was collected post event. The primary source of damage data for both events are Indonesian government post disaster damage reports (BAPPENAS 2006; BNBP 2009). This is supplemented by many smaller qualitative and quantitative studies undertaken after both events. The published report for the 2006 Yogyakarta earthquake is a preliminary report and updated damage figures that exceed those in the preliminary report are quoted in the Idham and Mohd (2018) study. It is these updated figures which are used in our analyses. The damage is reported in three damage states: heavy, moderate, and slight. The reports do not provide definitions for the damage states and the data is not disaggregated by building type. The damage is reported at the level of sub-province administrative divisions, known as *kabupaten* (district) or *kota* (city). These divisions differ in land area and population. The case study locations are shown in Fig. 1.

2.1 2006 Yogyakarta earthquake

The city of Yogyakarta is in the Special Region of Yogyakarta (DIY) in the south-central part of Java, between the southern coast of central Java and Mt Merapi to the north (Walter et al. 2008) (Fig. 1). Prior to the 2006 earthquake, the Opak Fault was regarded as the major active fault in the region and thought to be the source of damaging events in the past (Martin et al. 2022). The 27 May 2006 Mw 6.4 event occurred at 12 km depth on a previously unidentified fault, 20 km east of the Opak Fault (Walter et al. 2008). The 2006 event caused widespread damage in DIY and Central Java Province, with around 154 000 houses destroyed and 5716 fatalities (BAPPENAS 2006). To date this event is the most damaging earthquake to have impacted the island of Java (Kyaw et al. 2018; Walter et al. 2008). This event is one of the most well studied in Indonesia, however many studies are qualitative rather than quantitative (e.g., Boen 2006; Elnashai 2007; Rosyidi et al. 2008) or focus on the identification and characterization of the source and the ground motion (Pawirodikromo 2018; Tsuji et al. 2009; Walter et al. 2008) rather than the characterization of building fragility or vulnerability.

In the most severely impacted kabupaten of Bantul and Klaten, it is estimated that 90 and 60 percent of the houses were damaged, respectively, with over 30 percent of the houses in Bantul needing complete reconstruction. These kabupaten are some of the most densely populated rural kabupaten in Indonesia and the population is relatively poor. This observation is important as it is known that the poor are less likely to be able to afford seismically resilient houses (Rosyidi et al. 2008). Other impacted kabupaten range from the relatively wealthy and densely populated city known as Kota Yogyakarta, to the relatively poor and more sparsely populated rural regencies of Gunung Kidul and Kulon Progo (BAPPENAS 2006).

The most common house types that were present prior to the event were non-engineered low rise masonry houses. These houses can be divided into unreinforced masonry and confined masonry houses. Both types consist of clay brick walls and predominantly clay tile roofing. A feature of all of these houses is that there was often poor-quality materials and inadequate reinforcement (Boen 2006; Elnashai et al. 2007; Idham and Mohd 2018). Wood and bamboo frame houses were present in the impacted areas and performed better than the masonry houses (BAPPENAS 2006). The literature indicates that the same house types

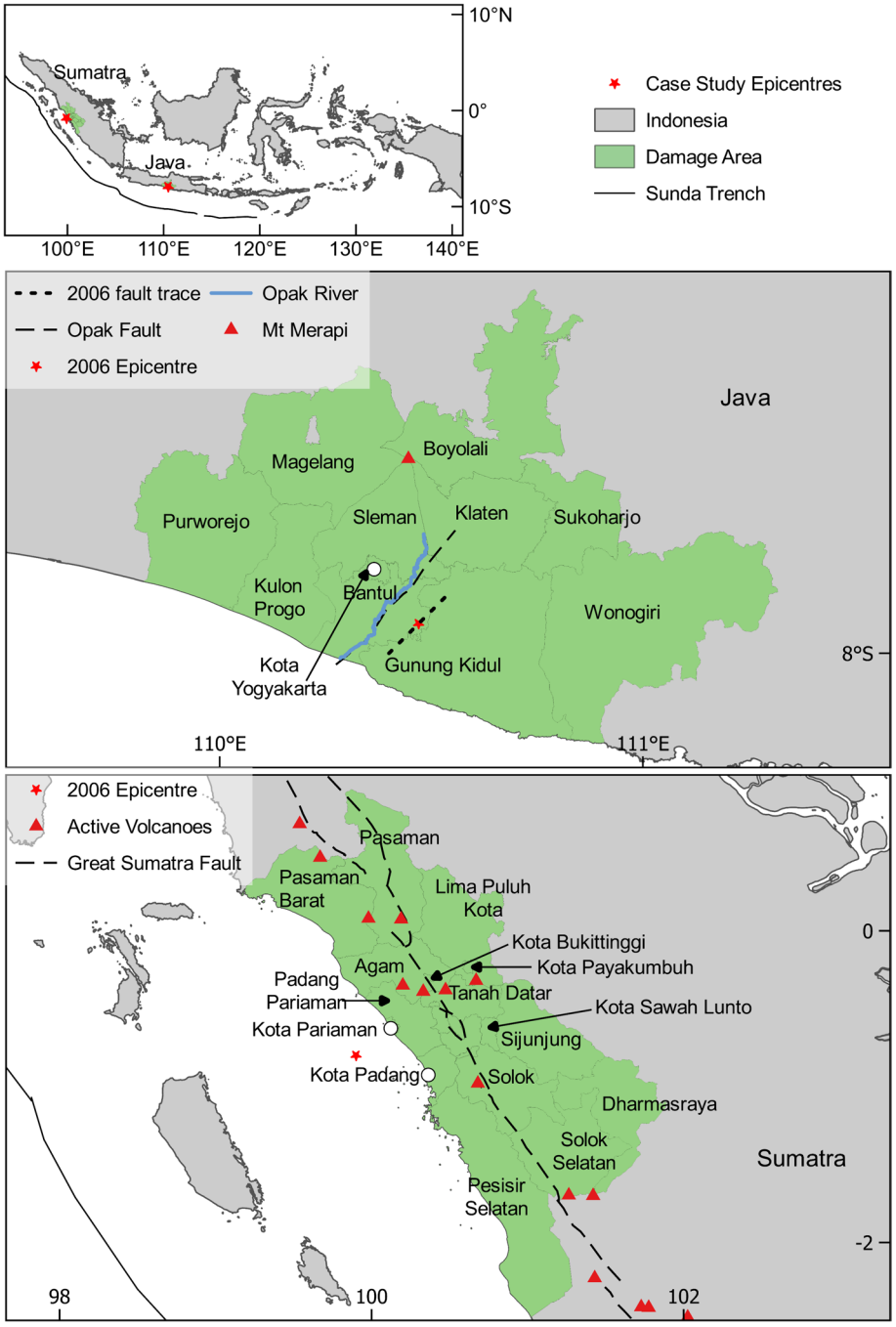


Fig. 1 Case study locations in Indonesia. The 2006 Yogyakarta earthquake (Java pane), an active shallow crust event that occurred 20 km to the east of the Opak Fault. Mt Merapi is the only active volcano in this area. The 2009 Padang earthquake occurred in West Sumatra (Sumatra pane), an intra-slab event on the Sunda Trench. There is a chain of active volcanoes that run parallel to the coastline and through the impacted kabupaten

are being constructed today, but in different proportions. Aulady and Fujimi (2019) have found that the more seismically resilient confined masonry is now the dominant housing type whereas prior to the 2006 earthquake it was unreinforced masonry (Elnashai et al. 2007), which is more fragile. Saputra et al. (2017) found that poor construction practices still persist particularly in low-income households. Idham and Mohd (2018) who surveyed the impacted regions find that the quality of new housing construction peaked after the event and has decreased with time. Thus, the findings of this study will still be relevant to the housing stock that is present today.

2.2 2009 Padang earthquake

Padang is located in the central part of the southwestern coast of the island of Sumatra, opposite the Sunda trench (Fig. 1). The 2009 Mw 7.6 earthquake was an intra-slab event that occurred offshore at 78 km depth to the northwest of Kota Padang. Widespread damage was recorded throughout West Sumatra Province and the shaking was felt in Jakarta (Putra et al. 2014). Around 115000 houses were destroyed, and there were 1100 fatalities (BNBP 2009). The severity of the event was exacerbated by landslides which destroyed three villages in the kabupaten of Padang Pariaman (BNBP 2009). There have been several studies undertaken of this event which have provided quantitative data on the types of houses present (Mulyani et al. 2015) and the damage sustained (Sengara et al. 2010), as well as qualitative damage surveys (Wilkinson et al. 2012). Post disaster surveys show that the most severely affected areas were Kota Padang, Kota Pariaman and Padang Pariaman kabupaten. These account for over 80% of the total damage to the housing sector (BNBP 2009).

The dominant housing type in the West Sumatra region was low rise masonry houses. As in Yogyakarta, the construction can also be characterized by non-engineered houses with poor quality materials and inadequate reinforcement (Chian et al. 2019). Of particular note in Kota Padang is the poor-quality clay brick, which was observed in some collapsed houses, where engineers found some bricks were so soft they could be crumbled by hand (Sengara et al. 2010). The housing sector in Kota Padang sustained the greatest proportion and severity of damage due to the ground shaking where seventy-one percent of the housing stock was damaged and over twenty percent needed complete reconstruction. Kota Padang and Kota Pariaman are cities that were some of the wealthiest locations in Indonesia at the time of the event, however they experienced the highest ground motion intensities and suffered the most damage (BNBP 2009). Other impacted kabupaten range from rural to urban and relatively wealthier to relatively poorer.

3 Hazard

Three components are required to simulate the ground motion intensity for scenario damage analysis: rupture model, ground motion model (GMM, also often referred to as Ground Motion Prediction Equations or GMPE) and Vs30 model. The key seismic parameters used to define the rupture models for these events are shown in Table 1. The fault source parameters for the Yogyakarta 2006 and Padang 2009 earthquakes were based on Walter et al. (2008) and Wiseman et al. (2012), respectively. The selection of Vs30 models and GMMs are discussed in further detail in the following sections.

Table 1 Fault Parameters used to define the simple fault source in OpenQuake for the 2006 Yogyakarta and 2009 Padang earthquakes

Parameter	Yogyakarta 2006	Padang 2009
Magnitude (Mw)	6.4	7.5
Rake (degrees)	0	138.0
Hypocentre (lat, lon)	− 7.93, 110.47	− 0.8, 99.9
Depth	12 km	78 km
Upper Seismic Depth	6 km	73 km
Lower Seismic Depth	18 km	87 km
Dip (degrees)	85.00	53.5
Fault Type	Simple fault geometry	Simple fault geometry

3.1 Vs30 model selection

Vs30 models are used to characterize the shear wave velocity in the top thirty meters of soil. The Vs30 values can be used to classify the soils according to the NEHRP scale (Dobry et al. 2000), and the classification or the actual value are used by GMMs to account for site conditions. Hazard studies commonly use the USGS Vs30 model if there are insufficient field Vs30 measurements to characterize the study area (Silva et al. 2020). The USGS Vs30 model is available for the globe and is based on topography derived from SRTM data (Allen and Wald 2007; Heath et al. 2020). In Japan an alternative approach has been developed which considers additional parameters to generate a Vs30 model (Matsuoka et al. 2006). Cipta et al. (2017) developed a Vs30 model for Indonesia by using the Matsuoka et al. (2006) equations, accounting for the geomorphology as well as slope angle.

Figure 2 shows the comparison of the USGS Vs30 model and the Cipta et al. (2017) model for the DIY and Central Java region. The USGS Vs30 model generally predicts harder soils than the Cipta et al. (2017) model. Of particular significance is the site classification in the most damaged kabupaten of Bantul and Klaten. Walter et al. (2008) attributes the damage pattern, particularly in Bantul and Klaten, to site amplification due to soft sediments in these locations. In both Bantul and Klaten the USGS model shows predominantly soft rock (site class C) whereas the Cipta et al. (2017) model shows predominantly stiff soil (site class D). This would indicate in these locations that the Cipta et al. (2017) model may be a better representation of the site conditions.

There is a small amount of field data available that can be compared to the Vs30 models. These studies have all used multi-channel analysis or surface waves to measure Vs30 (Le Ngal et al. 2019; Muzli et al. (p212: cited in Pawirodikromo 2018—original in Bahasa); Rahayu et al. 2016), however, the details of the field data are insufficient to determine whether they are representative of a single site measurement or a broader area. In general, the field data supports the presence of softer soils in Bantul and Klaten, which agrees better with post event observations and the Cipta et al. (2017) model. For this reason, the Cipta et al. (2017) model was selected. However, some simulations are undertaken with the USGS Vs30 model to determine the sensitivity to this selection.

To our knowledge there is no published Vs30 data for West Sumatra. However, post disaster reports comment on the soft soils particularly along the coast, and that this was a contributing factor to the level of damage that was experienced (Wilkinson et al. 2012). Figure 3 shows the comparison of the two Vs30 models for the impacted region. It is observed that the Cipta et al. (2017) model predicted softer soils across the whole region compared

with the USGS Vs30 model. Particularly in the coastal region, the softer soil of Cipta et al. (2017) is in better agreement with the alluvial soils present there. Therefore, we select the Cipta et al. (2017) Vs30 model in West Sumatra as well but undertake some simulations with the USGS Vs30 model to assess the sensitivity of the damage results to this selection.

3.2 Ground motion model (GMM) selection

GMMs are used to predict the ground motion in this study. Selection of GMMs in low data settings, where there are limited ground motion recordings available can be extremely difficult (Cotton et al. 2006; Ghasemi et al. 2020). The absence of localized GMMs or sufficient recordings to develop them has been highlighted as an issue in several Indonesia seismic hazard studies (Cipta et al. 2018, 2017; Griffin et al. 2019; Irsyam et al. 2020, 2010). Limited information is available on which GMM are best suited to use in Indonesia. There are no reliable ground motion records from these events and therefore it is not possible to select GMMs by agreement with station data nor is it possible to use station data for conditioning of the ground motion fields. Thus, several GMM have been selected for testing, which have firstly been compared to the known damage distribution to determine whether they are suitable. The use of multiple GMMs will allow us to assess the sensitivity of the damage simulation results to this input model selection.

3.2.1 Active shallow crust GMM (Yogyakarta earthquake)

There are many GMMs available for the active shallow crust tectonic setting that could be used for the Yogyakarta earthquake scenario. Several are stated to be globally applicable, however they have not been validated for use in Indonesia and no Indonesian ground motion recordings were used in the derivation of the models. The ground motion recordings used are dominated by records from North America (Boore et al. 2014; Campbell and Bozorgnia 2014; Chiou and Youngs 2014). Papua New Guinea (PNG) is located adjacent to Indonesia in a similar tectonic setting and thus GMMs applicable there may also be relevant to Indonesia. Therefore in this study we initially considered all of the active shallow crust GMMs used in both the Indonesia PSHA (Irsyam et al. 2020) and the PNG PSHA (Ghasemi et al. 2020). Therefore, we considered GMMs used in both the Indonesia PSHA (Irsyam et al. 2020) and the Papua New Guinea (PNG) PSHA (Ghasemi et al. 2020) for inclusion. PNG is located adjacent to Indonesia in a similar tectonic setting and thus GMM chosen there may also be relevant to Indonesia.

Some of the GMMs in the Indonesia PSHA have been superseded and thus were disregarded. The remaining active shallow crust GMMs were very similar and only one was chosen, the Chiou and Youngs (2014) GMM as it predicted a larger area with higher intensity ground motion, which better matched the observed damage distribution. The seismic hazard study for Papua New Guinea (PNG) of Ghasemi et al.(2020) uses the Cauzzi et al. (2014) and the Zhao et al. (2016a) GMMs, which are not included in the Indonesia PSHA. The Cauzzi et al. (2014) is a global GMM but includes many records from Japan in its derivation, and the Zhao et al. (2016a) GMM was developed for Japan using entirely Japanese ground motion recordings. An earlier iteration of this GMM (Zhao et al. 2006) was found suitable for the Philippines (Naguit et al. 2017), which is located near Indonesia in a similar tectonic setting. We hypothesize that Indonesia is similar to Japan due to both being island archipelagos with active volcanism, rather than continental USA. Therefore, we have also selected for testing the Chiou and Youngs (2014) with parameters adjusted for Japan.

Fig. 2 The top pane is the USGS Vs30 Model for the DI Yogyakarta and Central Java Provinces impacted by the 2006 Yogyakarta earthquake. The NEHRP soil classification in the region ranges from B to E and is predominantly class C and D. The bottom pane shows the Cipta et al. (2017) Vs30 model for the region. The NEHRP soil classification in the region ranges from E to C and is predominantly class D

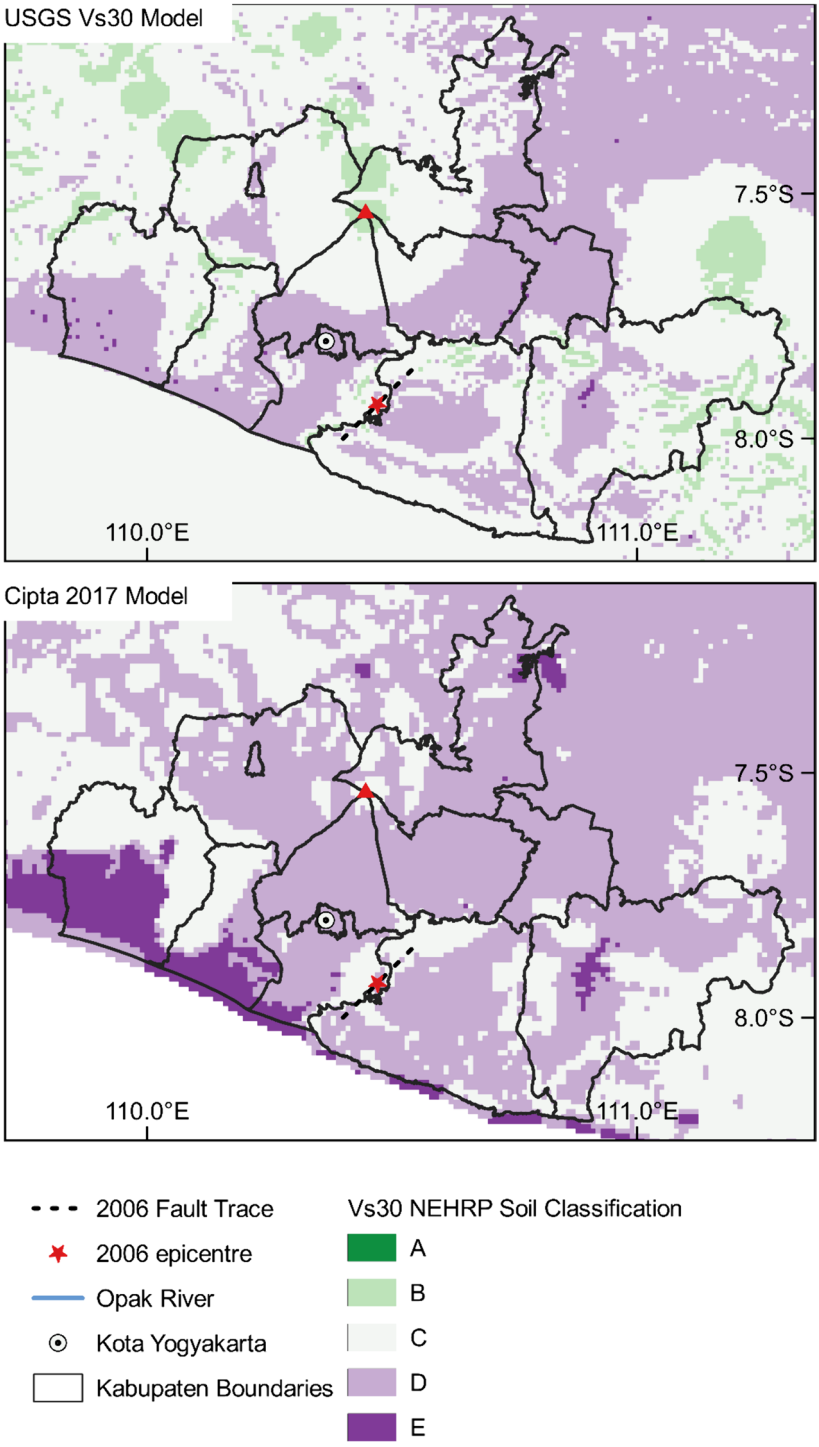
This GMM is designed to account for the known differences in site conditions and anelastic attenuation of seismic waves that occur in Japan. The ground motion for the impacted regions in DIY and Central Java has been modelled using the OpenQuake engine using these four GMMs – that is; Chiou and Youngs (2014) (both global and Japan variations), Zhao et al. (2016a), and, Cauzzi et al. (2014) – with the Cipta et al. (2017) Vs30 model. The results are shown in Fig. 4.

Figure 4 shows a significant degree of variability in the maximum PGA values simulated by the four GMMs selected. All GMMs predict ground motion that is broadly consistent with the overall spatial pattern of observed damage, with higher PGA in the most impacted areas of Bantul and Klaten and lower in areas with less damage. The exception to this for all GMM are the high PGA values in Gunung Kidul relative to Bantul, which is not consistent with the pattern of observed damage in these kabupaten. The Cauzzi et al. (2014) appears to match better with damage observations in Gunung Kidul and in the more distal locations, whereas the Zhao et al. (2016a) appears to be more reflective of the damage observed in Bantul. There is no clear GMM which best matches the observed damage and therefore damage simulations are conducted with all four GMM.

3.2.2 Intra-slab GMMs (Padang earthquake)

For the intra-slab Padang event there are fewer GMMs available. None of these models include Indonesian ground motion recordings in their derivations, though they do include recordings from most other subduction zone locations. We chose to model the ground motion using the Atkinson and Boore (2003) GMM that is included in the Indonesia PSHA of Irsyam et al. (2020), the Zhao et al. (2016b) and Abrahamson et al. (2016) GMMs that are used in the PNG PSHA of Ghasemi et al. (2020), and the Atkinson and Boore (2003) with the parameters adapted for Japan. The reasoning for this is the same as for the active shallow crust GMMs, that Indonesia is likely to be more similar to Japan than to continental subduction zones.

The damage distribution showed a significant decrease in damage sustained in kabupaten located within or beyond the active volcanic zones that run parallel to the coast and the Sumatran Fault (see Fig. 5). We speculate this is because of the increase in anelastic attenuation that is observed when seismic waves cross volcanic zones, as has been observed in Japan by Abrahamson et al. (2016) and Zhao et al. (2016b). They have developed parameters in their GMM to account for this phenomenon. The Abrahamson et al. (2016) GMM allows sites to be assigned as fore-arc or back-arc and uses a much steeper attenuation rate for back-arc sites. Thus, the Abrahamson et al. (2016) GMM with a defined back-arc region is considered in this study. The fore-arc and back-arc regions are defined using the zones in Mukti et al. (2021). Zhao et al. (2016b) use a parameter called *Rvolc*, which is the distance on the surface of the earth that the seismic waves travel through a defined volcanic zone. This is then used to adapt the attenuation relationship in the GMM. Unfortunately, the Rvolc parameter is not yet implemented in practice in the OpenQuake engine and thus it has not been used in this study.


USGS Vs30 Model
7.5°S
8.0°S
110.0°E
111.0°E
Cipta 2017 Model
7.5°S
8.0°S
110.0°E
111.0°E
2006 Fault Trace
2006 epicentre
Opak River
Kota Yogyakarta
Kabupaten Boundaries
Vs30 NEHRP Soil Classification
A
B
C
D
E

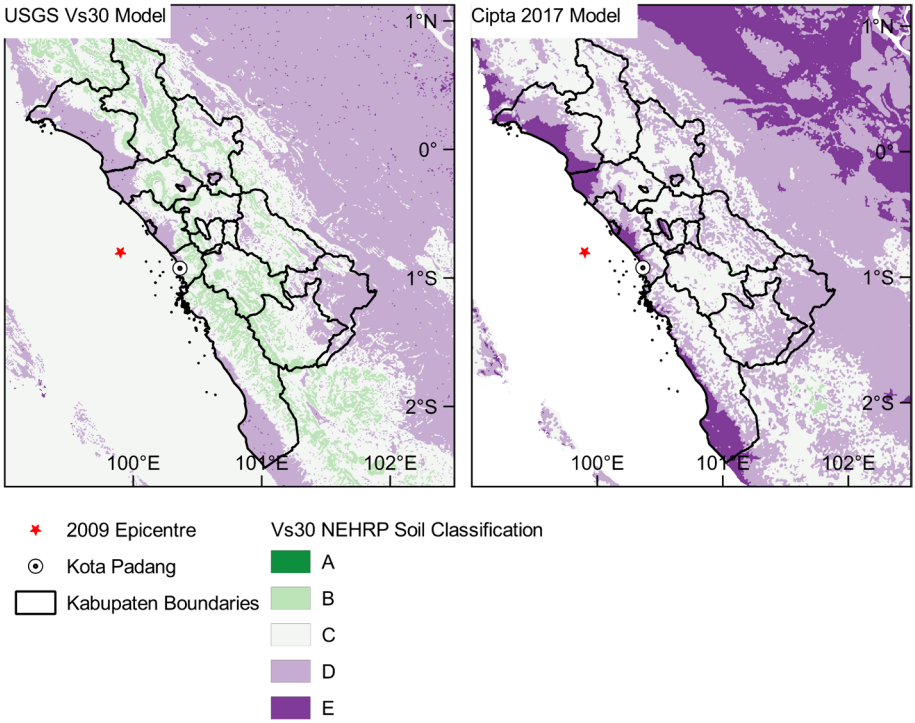


Fig. 3 The pane on the left shows the USGS Vs30 model for the West Sumatra province that was impacted by the 2009 Padang earthquake. The NEHRP soil class in the region ranges from site B to site E, with the impacted region predominantly class B and C. The pane on the right shows the Cipta et al. (2017) Vs30 model for the same region. The NEHRP soil classification ranges from B to E, with the impacted region predominantly class C and D

Figure 5 shows the average PGA simulated for West Sumatra for the 2009 Padang event. The only GMM that broadly agrees with the damage distribution in the volcanic regions and beyond is the Abrahamson et al. (2016). In the fore-arc region the simulated ground motion using the Youngs et al. (1997) and Abrahamson et al., (2016) are similar. The other GMM all predict a much larger area of higher intensity ground motion that is inconsistent with the observed damage pattern. Therefore, the Abrahamson et al. (2016) GMM is used for all damage simulations for this event.

4 Residential exposure

An exposure model was developed for each location that represents the housing stock present in the year before each event occurred. The exposure models were developed using the broad steps outlined by Yepes-Estrada et al. (2017): identification of the building classes and mapping the building classes to the census. Finally, the buildings were then spatially distributed on a 500 m grid using population density to enhance the accuracy of the model (Dabbeek et al. 2021). The details for each step are described in the following sections.

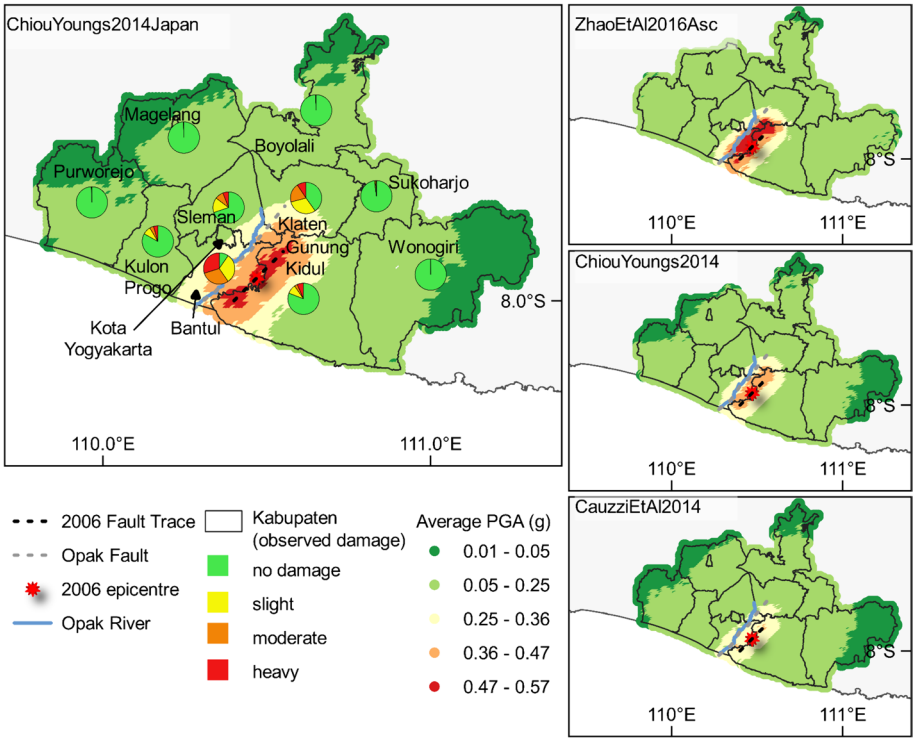


Fig. 4 The four GMMs considered in this study to model the ground motion from the 2006 Yogyakarta earthquake. The surface ground motion shown is the average PGA (g) from 1000 ground motion simulations conducted using the OpenQuake Engine (version 3.11). The observed damage to the housing stock is overlaid to show the agreement between the ground motion and the damage pattern

4.1 Residential building classes

Due to the number of post disaster studies the types of houses that were present prior to each event are quite well known. This information has been supplemented with studies on housing in general in Indonesia to characterize the houses using the GEM taxonomy (Brzev et al. 2013). This allows for the consistent description of the major materials of construction and the lateral load resisting system to ensure that they can be easily associated with both the census data and fragility functions for damage assessment. The building types that have been identified are shown in Table 2. All the sources used in this study indicate that one story houses are dominant in both Padang and Yogyakarta. Thus, all houses are considered to be one story for simplicity, unless additional information was available for a specific building type.

4.2 Building classes from census data

In Indonesia there is census data reported at the kabupaten level which includes information on the main wall, floor, and roof material from 1980 to 2000 (IPUMS 2020a, 2020b,

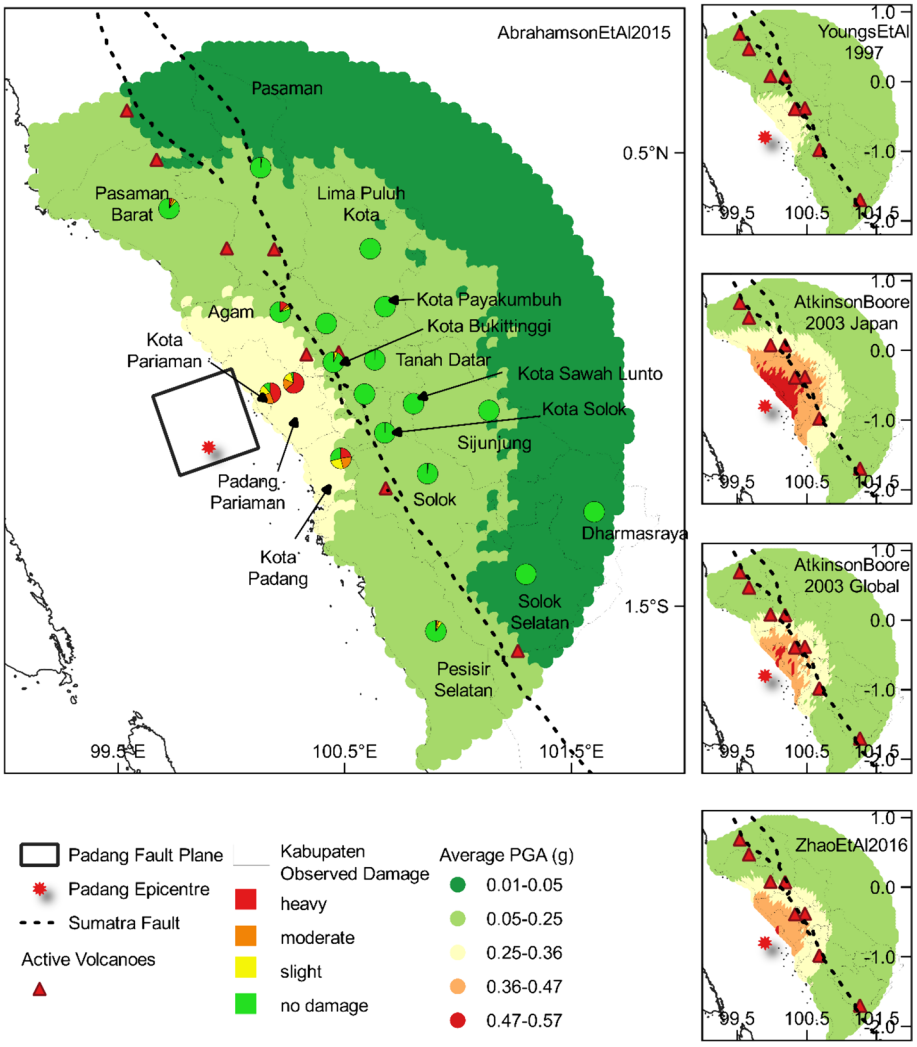


Fig. 5 The five GMMs that were considered in this study for the surface ground motion from the 2009 Padang earthquake. The location of the Great Sumatran Fault Zone and the chain of active volcanoes is also shown. The volcanoes are important to understand the damage distribution due to this event

2020c) as well as the number of houses. There are supplementary smaller surveys in the intervening years which contain the same information reported at the same administrative level (BPS 2005, 2009; IPUMS 2020d). The multiple eras of data have been used to estimate building age, which is critical to allow mapping to the types of houses identified in Table 2. The census data confirms that low level houses are the dominant type of housing in both study areas. It also shows the variation in construction materials between rural and urban locations within each kabupaten. Table 3 summarizes the data obtained from the census datasets to characterize the houses.

Table 2 Building classes identified in Yogyakarta and Padang at the time of the events (compiled from: BAPPENAS 2006; BNBP 2009; Boen 2006; Sengara et al. 2010; Idham and Mohd 2018; Teguh 2014)

Type	Description	Age	Taxonomy*
Unreinforced masonry, Dutch era  (Image: CEphoto, Uwe Aranas[a])	Thick clay fired brick walls (~100 mm thick/high bricks), clay tile roof. Originally lime washed on the outside. Now very poorly maintained with crumbling mortar	Pre 1980	MUR-CL99_LWAL-DNO_H:1
Unreinforced masonry, modern  (Image: Padang Survey Team, Sengara et al. 2010)	Thin clay fired brick walls (~50 mm thick/high bricks), generally with poor quality mortar (commonly 1 part lime and 2 parts sand), no reinforcements, clay tile roof. Often rendered on the outside	Post 1980	MUR-CL99_LWAL-DNO-H:1
Unreinforced masonry (Padang Special) No image available	A type of unreinforced masonry house only found in Padang. 700 mm tall dado masonry wall, timber posts, coarse chain wire mesh between posts plastered with cement stucco	All ages	MUR-CL99_LWAL-DNO-H:1
Confined masonry  (Image: Phil Cummins)	Thin clay fired brick walls, with varying quality. Lightly reinforced concrete columns and beams	Post 1980, generally post 1990	MCF_LWAL-DUL_H:1 MCF_LWAL-DUM_H:1

Table 2 (continued)

Type	Description	Age	Taxonomy*
Wood -traditional  (Image: Phil Cummins)	Traditional style wood construction often with heavy wooden roofing or clay tile roofing. High quality workmanship	All ages	W_LFM-DUM_H:1
Bamboo  (Image: CEphoto, Uwe Aranas[b])	Light wood or bamboo frames with light bamboo cladding. Roof varies. Generally two story houses. Considered semi-permanent and informal housing	All ages	W_LFM-DUL_H:2
Wood-modern  (Image: Padang Survey Team, Sengara et al. 2010)	Wood housing with light frames and light cladding. Roofing varies. Generally perform well	All ages	W_LFM-DUM_H:1

Table 2 (continued)

Type	Description	Age	Taxonomy*
Reinforced concrete frame with clay brick infill  (Image: Phil Cummins)	Standard brick walls (65 mm thick bricks) with a clay tile roof. Materials of varying quality. Noted in Padang but not in Yogyakarta. Commonly three-story houses	Post 2000	CR_LFINF-DUL_H:3

*MUR = unreinforced masonry, MCF = confined masonry, DNO = no ductility, DUL = low ductility, DUM = moderate ductility, LWAL = wall, LFM = frame, H = height, CL99 = clay brick unknown type, W = wood, INF = infilled walls, CR = reinforced concrete. [a]https://commons.wikimedia.org/wiki/File:Yogyakarta_Indonesia_Colonial-houses-01.jpg

[b] https://commons.wikimedia.org/wiki/File:Yogyakarta_Indonesia_Buildings-in-Yogyakarta-01.jpg

As in other parts of the world (e.g. South-America; Yepes-Estrada et al. 2017) the census data is not in itself adequate to allow correlation with the building classes that were present (Table 2). Houses listed in the Census as having masonry walls for example could actually be one of several masonry building types identified in Table 2, and therefore a mapping scheme was developed using other indicators and expert opinion to link the Census data to the known buildings present.

4.3 Number and type of houses

The number of buildings in the census have been mapped to the types of buildings present using a combination of the following factors: age, wall material, setting (city, urban, rural) and relative poverty (see Table 4). The poverty decile ranking has been used to determine the relative poverty of regions. It is a ranking of all the kabupaten in Indonesia, based on the percentage of people who live below the poverty line. This ranking is reported in the post-disaster reports for all kabupaten impacted by the Yogyakarta and Padang earthquakes (BAPPENAS 2006; BNBP 2009).

These factors are used to define the quality of the structures. For example, the poorest urban areas will likely continue to use unreinforced masonry for longer, as was observed in Bantul and Klaten (Elnashai et al. 2007; Rosyidi et al. 2008), while the wealthiest regions will likely introduce confined masonry earlier and construct with higher quality materials. This is reflective of observations that the poor are the most likely to have poor seismic provisions (BAPPENAS 2006; BNBP 2009; Rosyidi et al. 2008), including poor quality or a complete absence of seismic reinforcing elements and therefore low or no ductility of structures. Martins and Silva (2020) use the age and whether construction is informal or not to assign ductility. Thus we have used the poverty decile, the age, and the setting as indicators of the type of masonry construction and the level of ductility of the houses. As most houses in Indonesia at the time of the events were either informal (bamboo houses) or non-engineered, this results in the ductility of houses ranging from non-ductile to moderate ductility. The types and number of buildings were then adjusted if additional information were available, for example in Bantul it is known that ~ 95% of the buildings were unreinforced masonry (Elnashai et al. 2007) due to its Dutch Colonial history.

Table 3 Building information reported in the Census, SUSENAS and Inter-Census surveys

Data Set	Wall material	Age range	Height	Location
1980 Census	Masonry, wood, bamboo	Pre 1980	> 95% 1 story	Urban/rural
1990 Census	Masonry, wood, bamboo	1980–1990	> 95% 1 story	Urban/rural
2000 Census	Masonry, wood, bamboo	1990–2000	NA	Urban/rural
2005 Inter Census Survey	Masonry, wood, bamboo	Post 2000	> 95% 1–2 story houses (not separated out in census)	Urban/rural
2009 Susenas (Household) Survey	Masonry, wood, bamboo	Post 2000	> 95% 1–2 story houses (not separated out in census)	Urban/rural

Table 4 Mapping Scheme to link the census building types to the observed building type using the GEM taxonomy (Brzev et al. 2013) for ductility and seismic resisting elements

Census Building Type	Setting (urban/rural/city)	Poverty Decile Ranking (1 poorest, 10 wealthiest)	Ductility	Type	Taxonomy from Table 2
Masonry pre-1980	All	1 to 7	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		> 7	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
Masonry 1980–1990	Rural/Urban	1 to 7	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		> 7	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
	City	1 to 4	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		> 4	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
Masonry 1990–2000	Rural/City	1 to 4	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		> 4	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
	Urban	1 to 7	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		> 7	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
Masonry 2000–2005	Rural	All	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
	Urban	1 to 7	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
		> 7	Moderate ductility	Confined masonry	MCF_LWAL-DUM_H:1
	City	1 to 4	Non-ductile	Unreinforced masonry	MUR-CL99_LWAL-DNO_H:1
		5 to 6	Low ductility	Confined masonry	MCF_LWAL-DUL_H:1
		> 6	Moderate ductility	Confined masonry	MCF_LWAL-DUM_H:1
Wood	All	All	Moderate ductility	Wood	W_LFM-DUM_H:1
Bamboo	All	All	Low ductility	Wood	W_LFM-DUL_H:2

The proportions of building types in each Province, based on the implementation of the mapping scheme, are shown in Fig. 6. Thus in the exposure model DIY has a higher proportion of unreinforced masonry houses than either Central Java or West Sumatra. This is due to age of the building stock in DIY and also the relatively poorer population. Conversely, we see that there is a much higher proportion of confined masonry houses in Central Java and West Sumatra which is influenced by both the relatively higher wealth of these locations and the younger building stock. The significantly higher proportion of traditional wood houses (Wood-DUM) in West Sumatra is contributed to by the relative wealth of the region and the many rural kabupaten with low population density. In general, the exposure model assumes that poorer regions have more fragile houses and wealthier areas have less fragile houses, densely populated areas have more fragile houses and more rural areas have less fragile houses. The resulting proportions of house types in the exposure model is consistent with the limited field data from both locations (BAPPENAS 2006; BNPB 2009; Chian et al. 2009; Elnashai et al. 2007; Sengara et al. 2010).

4.4 Spatial distribution of buildings

The classes and proportions of houses present has thus far been discussed at the kabupaten level. The number of houses needs to be distributed spatially to give a more realistic model for the maximum ground motion experienced by each house. It is an accepted methodology to use population models to distribute the houses and this has been shown to increase the accuracy of the model (Dabbeek et al. 2021). In this study we use the WorldPop 2005 model (WorldPop 2005) with the population per pixel at 500 m resolution to distribute the houses for Yogyakarta and Central Java as this is just prior to the event. This data is then normalized to the known number of houses in each kabupaten from the 2005 Inter-census survey, to ensure that the total number of houses in the exposure model is accurate. For Padang we use the WorldPop 2010 model (WorldPop 2010) to distribute the houses. Unfortunately, this is a dataset which post-dates the 2009 event. However, this is normalized to the number of known houses in each kabupaten from the 2009 SUSENAS (BPS 2009), which reduces this potential source of bias.

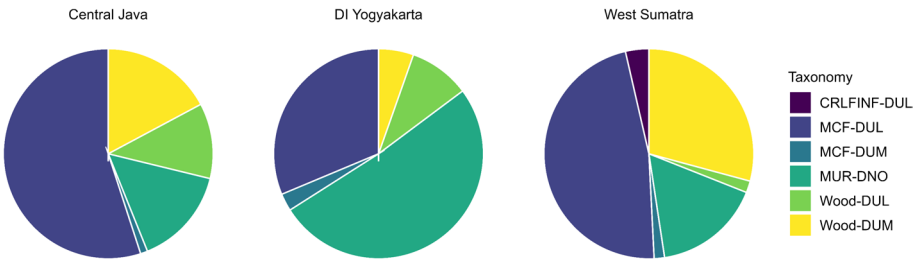


Fig. 6 The proportions of housing types for each impacted Province in this study. (MUR-DNO = unreinforced masonry with no ductility, MCF-DUL: Confined masonry with low ductility, MCF-DUM = confined masonry with moderate ductility, CRLFINF-DUL = reinforced concrete frame with brick infill walls and low ductility, Wood-DUL = light wood or bamboo frame and light wood or bamboo cladding with low ductility, Wood-DUM = light wood frame and light wood cladding with moderate ductility)

4.5 Uncertainty in exposure models

There is an unknown degree of uncertainty in exposure models developed from multiple secondary data sources. Much of this uncertainty is introduced by using data that was not originally collected for this purpose. Thus, the uncertainty in the exposure models developed for this study is unknown. In this study there is confidence in the number of houses and the spatial distribution at the kabupaten level, as this information is available in the census data. There is also confidence in the types of houses present and the materials of construction due to the large amount of post disaster studies. The main uncertainty is associated with the proportions of each type of house and the spatial distribution of these buildings at the 500 m scale. As the events are historical and the built environment has inevitably changed it is not possible to reduce the uncertainty further with ground surveys. The uncertainty in the exposure model and the impact on the final damage simulations is discussed in detail in the results section.

5 Performance of structures and fragility functions

5.1 Observed performance

The maximum PGA experienced during the 2006 Yogyakarta earthquake is estimated to be 0.45 g 0.5 g (Pawirodikromo 2018, 2020). This is supported by the ground motion modelling discussed in Sect. 3. However, much of the impacted area likely experienced a maximum PGA significantly below this. Pawirodikromo (2020) suggest that the maximum PGA was 0.45 g in Bantul, and not at the epicenter. Further to the west Elnashai (2007) used the only available partial station records to estimate the maximum PGA in Kota Yogyakarta as between 0.19 g and 0.33 g. Given the relatively modest maximum PGA and the pervasive heavy damage, performance of these houses appears to have been poor. As discussed above, this has been attributed to poor construction, lack of maintenance and the absence of seismic provisions in the residential building stock (Boen 2006; Bothara et al. 2010; EERI 2006, 2009; Rosyidi et al. 2008). The preliminary damage report from the 2006 Yogyakarta event (BAPPENAS 2006) indicate that wood structures outperformed the masonry structures, with significantly smaller proportions collapsing or sustaining damage.

In the Padang 2009 earthquake the estimate of the maximum PGA experienced is 0.44 g (Sengara et al. 2010). This is supported by the ground motion modelling discussed in Sect. 3. There is no strong motion data available near the source (Sengara et al. 2010), but the ground motion modelling in this study and that undertaken by Sengara et al. (2010) indicates that much of the region experienced ground motion below this value. As was observed in Yogyakarta, most of the damage was sustained in masonry houses that were of poor quality (BNBP 2009). The next step is to determine how the fragility of these buildings can be best represented with existing fragility functions.

5.2 Fragility functions

There are empirical fragility functions that were derived from a small damage survey after the Padang 2009 earthquake (Sengara et al. 2010). However, these functions are derived in MMI and only two MMI values were recorded, thus they are not expected to

be representative of a wide range of ground motion values. Therefore, in this study we test several functions to determine which perform best in terms of estimating the damage that was observed after the case study earthquakes. Martins and Silva (2020) have developed globally applicable functions for the main building types in the world to address the absence in fragility functions for many countries. Therefore, these functions were selected for this study. The Villar-Vega et al. (2017) functions that were developed for use in South America were also selected due to the similarities in the descriptions of construction styles and materials of non-engineered houses in South America and in Indonesia. Both the Martins and Silva (2020) and the Villar-Vega et al. (2017) functions are analytically derived functions, and for the types of buildings in Indonesia the functions are quite similar. In addition, the widely used HAZUS (FEMA 2020) functions were selected as they have been used in previous studies in Indonesia (Pratiwi et al. 2019). The HAZUS functions are derived through a combination of analytical methods, expert opinion, and empirical data. The HAZUS functions exist for various code levels in the USA, the code level has been selected based on the building description that best matches our knowledge of the building types in Indonesia. There are no confined masonry functions in HAZUS. However in the Padang damage survey (Sengara et al. 2010) confined masonry and low-rise reinforced concrete with brick infill walls were found to have almost comparable performance. Thus the decision was taken to apply the HAZUS fragility functions for reinforced concrete with infill walls to confined masonry houses in this study. The functions for a one-story unreinforced masonry house from each of these studies is shown in Fig. 7 and the correlation between functions and housing types for Indonesia are shown in Table 5. For comparison

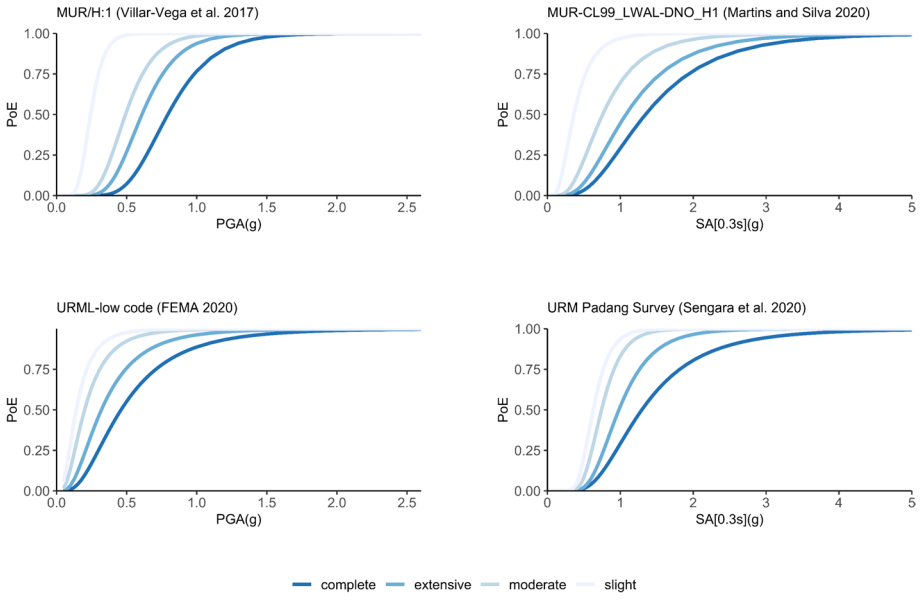


Fig. 7 The fragility functions considered in this study for low rise unreinforced masonry houses with clay bricks. The Martins and Silva (2020), Villar-Vega et al. (2017) and FEMA (2020) functions have four damage states: slight, moderate, extensive, and complete. Martins and Silva (2020) use spectral acceleration (0.3 s) for the ground motion intensity whereas the Villar-Vega et al. (2017) and the FEMA (2020) use PGA (g). The Sengara et al. (2010) function that was developed in Padang is shown for comparison. This function also has four damage states and a ground motion intensity in SA(0.3 s)

Table 5 Building types in the exposure model and the corresponding fragility function in each of the fragility function datasets

Taxonomy (linked to Table 2 and Table 4)	HAZUS (FEMA 2020)	Global (Martins and Silva 2020)	South America (Villar-Vega et al. 2017)
MUR-CL99_LWAL_DNO_H:1 (one story unreinforced masonry clay brick)	URML–low code	MUR-CL99_LWAL-DNO_H:1	MUR/H:1
MCF_LWAL_DUL_H:1 (one story confined masonry with low ductility)	C3L–pre-code	MCF_LWAL_DUL_H:1	MCF/H:1/DNO
MCF_LWAL_DUM_H:1 (One story confined masonry with moderate ductility)	C3L–low code	MCF_LWAL_DUM_H:1	MCF/H:1
W_LFM_DUL_H:2 (two story wood frame with light wood cladding, moment frame, low ductility)	W1 low rise	W_LFM_DUL_H:2	W + LFM + DUL/H:2
W_LFM_DUM_H:1 (One story wood frame with light wood cladding, moment frame, with moderate ductility)	W1 low rise	W_LFM_DUM_H:1	W + LFM + DUM/H:1
CR_LFINF_DUL_H:3 (Three story reinforced concrete frame with infill walls and low ductility)	C3L- low code	CR_LFINF_DUL_H:3	CR/LFINF + DNO/H:3

the fragility function derived from the Padang study for unreinforced masonry houses (Sengara et al. 2010) has been converted to SA (0.3 s) and is also shown in Fig. 7. Figure 7 highlights that there can be great variation in existing fragility functions for similar building types.

Selecting representative functions from existing datasets is challenging due to the limited information available on the characterization of the buildings that fragility functions represent. Therefore, all fragility functions in Table 5 are tested to determine what the sensitivity to this input model selection is and which ones are best suited for use in Indonesia.

5.2.1 Damage states

To allow comparison of the damage simulation results with different fragility functions and observed damage data, the damage states used need to be harmonized. In Indonesia damage data are collected and reported in three damage states: heavy, moderate, and slight. Commonly fragility functions use four damage states, but this is not always the case. Table 6 shows the damage states and descriptions for each set of fragility functions used in this study and how they align with the Indonesian damage data. The Villar-Vega et al. (2017) functions are not documented in the table as they are defined by spectral displacement yield point and this is not translated to a damage description. The harmonization described in Table 6 is used in the analysis of the results of the damage simulations.

6 Damage simulation method and results

Damage simulations were undertaken for the Yogyakarta and Padang case studies to compare the simulated results with the observed damage data reported at the kabupaten level. Damage simulations were conducted using the OpenQuake version 3.11 scenario damage calculator (Silva et al. 2014) for each combination of GMM and fragility function set for each case study. Each damage simulation consists of 1000 ground motion realizations, calculated in both PGA and SA (0.3 s), to achieve convergence (Silva 2019) and the average damage calculated from these realizations. Further simulations were undertaken to refine the fragility functions used based on the initial results, and to examine the impact of Vs30 model selection. Twenty-one damage simulations were undertaken for the Yogyakarta case study and six for the Padang case study. The results allow us to draw conclusions about the impact of input model selection on the accuracy of damage simulations and to make recommendations for future seismic damage and risk assessment in Indonesia.

6.1 Input model selection sensitivity

The influence of fragility model selection was assessed at a high level by examining the percentage of housing stock damaged in simulations using different fragility models. In Table 7 we list the range in percentage of predicted to sustain any level of damage by kabupaten, for each of the four GMMs considered. In each column of Table 7 the GMM is held constant and the differences in percentage damage result from varying the choice of fragility functions over those indicated in Table 6.

Table 8 shows the same data when the fragility functions are held constant and the GMM is varied. Table 7 shows that changes in fragility function selection can cause the total damage results to differ by up to 60% in some kabupaten. Whereas Table 8 shows us

Table 6 Descriptions of damage States in each of the fragility functions datasets and the Indonesia post-event damage reporting

Damage state	HAZUS (FEMA 2020)	Global (Martins and Silva 2020)	Indonesia (BNBP 2009; Idham and Mohd 2018)
Slight	Diagonal, stair-step hairline cracks on masonry wall surfaces; larger cracks around door and window openings in walls with a large proportion of openings; movements of lintels; cracks at the base of parapets	0.05 loss ratio (ratio of repair cost to replacement cost) 0.3 CoV (coefficient of variation)	Cracks in non-structural members. Habitable
Moderate	Most wall surfaces exhibit diagonal cracks; some of the walls exhibit larger diagonal cracks; masonry walls may have visible separation from diaphragms; significant cracking of parapets; some masonry may fall from walls or parapets	0.20 loss ratio 0.20 CoV	Cracks on structural members
Extensive	Most walls have suffered extensive cracking. Some parapets and gable end walls have fallen. Beams and trusses may have moved relative to their supports	0.60 loss ratio 0.10 CoV	Termed Heavy damage—reconstruction required
Complete	Structure has collapsed or is in imminent danger of collapse due to in-plane or out-of-plane failure of walls	1.00 loss ratio 0.00 CoV	Termed Heavy damage—reconstruction required

Table 7 The sensitivity of damage simulation results for kabupaten in DIY and Central Java to the choice of fragility functions. Each column corresponds to a set of simulations using a fixed GMM, and the range of percentage damage reflects the variation in results due to different choice of fragility functions (i.e., for each GMM there are three damage simulations, one for each set of fragility functions)

Kabupaten	Ground motion model (range in % housing stock predicted to be damaged)			
	Cauzzi et al. (2014)	Chiou and Youngs (2014)	Chiou and Youngs (2014) -Japan	Zhao et al. (2016a)
Bantul	35.8	43.8	41.8	36.8
Boyolali	19.8	21.7	22.1	27.9
Gunung Kidul	38.4	42.2	43.0	37.2
Klaten	41.7	49.9	51.2	46.2
Kota Yogyakarta	50.9	60.9	63.5	56.3
Kulon Progo	30.8	35.5	39.1	37.4
Magelang	16.3	17.3	16.1	27.6
Purworejo	18.3	20.9	22.0	24.8
Sleman	44.1	53.0	55.0	50.5
Sukoharjo	36.5	42.6	45.7	43.3
Wonogiri	20.8	22.7	22.2	31.4

that GMM selection has significantly less impact on the total damage results, where the results differ by up to 15%. This suggests that the variability in damage simulation results is to first order controlled by the fragility function selected.

As the Abrahamson et al. (2016) GMM with defined fore-arc and back-arc regions was the only GMM used for the 2009 Padang case study, there is no analysis of the sensitivity

Table 8 The sensitivity of damage simulation results for kabupaten in DIY and Central Java to the choice of ground motion model. Each column corresponds to a set of simulations using a fixed fragility function, and the range of percentage damage reflects the variation in results due to different choice of ground motion models (i.e., for each set of fragility functions there are four damage simulations, one for each ground motion model)

Kabupaten	Fragility Function (range in % housing stock predicted to be damaged)		
	HAZUS (FEMA 2020)	Global (Martins and Silva 2020)	South America (Villar-Vega et al. 2017)
Bantul	13.5	8.6	14.3
Boyolali	9.2	1.3	1.7
Gunung Kidul	11.6	11.6	13.1
Klaten	12.2	5.6	6.8
Kota Yogyakarta	13.7	4.9	6.3
Kulon Progo	8.6	2.5	4.4
Magelang	13.0	1.2	1.5
Purworejo	6.5	0.1	0.4
Sleman	11.9	5.2	6.7
Sukoharjo	8.6	0.9	1.7
Wonogiri	11.2	0.8	1.1

of results to the GMM selection. However, it is noted that if other GMMs had been selected that do not account for the increased inelastic attenuation due to the presence of a chain of active volcanoes, the damage simulations would show significant damage to the east of the volcanoes, where none was observed. Table 9 shows the sensitivity of the results to the selection of the fragility functions. As seen in the Yogyakarta simulation results, there is a wide range in the simulated percentage of houses that experienced damage depending on which fragility functions are chosen. At this high level the results show, as expected, that seismic damage simulations are very sensitive to the selection of fragility functions, and in Yogyakarta and Padang this means simulations can vary by over 60% in the predicted number of houses damaged in a damage category. This can lead to both gross underestimation and overestimation depending on the location and the functions selected.

The sensitivity to the Vs30 model selection was also tested. Table 10 shows the example of the variation in the results for the 2009 Padang event using the HAZUS fragility and both the USGS Vs30 model and the Cipta Vs30 model. The HAZUS functions were chosen as they are most sensitive to small changes in the predicted ground motion when the ground motion intensity is low. The results show that there is very small variation in the overall predicted damage of the simulations. Similar results are found with comparisons in Yogyakarta as well. Thus, we conclude that the damage simulation results at the kabupaten scale are less sensitive to Vs30 model selection than the fragility function and GMM

Table 9 Comparison of the observed percentage of houses damaged by the 2009 Padang earthquake and the range of damage predicted in the damage simulations. This is the results of four damage simulations with the Abrahamson et al. (2016) GMM and the three sets of fragility functions in this study

Kabupaten	Percentage of houses damaged			
	Observed (BNPB 2009)	Simulation minimum	Simulation maximum	Simulation range
Agam	20.4	17.4	63.8	46.4
Dharmasraya	0.0	0.1	4.1	4.0
Kota Bukittinggi	0.0	9.7	63.5	53.9
Kota Padang	71.2	30.9	79.8	48.9
Kota Padang Panjang	4.9	12.6	73.0	60.4
Kota Pariaman	71.3	44.9	79.6	34.7
Kota Payakumbuh	0.0	3.6	43.7	40.2
Kota Sawah Lunto	0.0	5.2	51.4	46.1
Kota Solok	0.1	8.0	60.9	52.9
Lima Puluh Kota	0.0	3.5	36.2	32.7
Padang Pariaman	95.7	33.2	80.8	47.6
Pasaman	2.1	3.3	27.7	24.3
Pasaman Barat	12.1	11.3	53.3	42.0
Pesisir Selatan	10.0	8.4	43.9	35.5
Sijunjung	0.0	1.9	26.0	24.1
Solok	0.9	11.3	47.8	36.5
Solok Selatan	0.0	4.0	26.7	22.7
Tanah Datar	0.3	9.5	58.7	49.2

Table 10 Comparison of the observed percentage of houses damaged by the 2009 Padang earthquake using the HAZUS fragility functions and either the Cipta et al. (2017) Vs30 model or the USGS Vs30 model. Both simulations were undertaken using the Abrahamson et al. (2016) GMM

	Observed (BNPB 2009)	Cipta Model	USGS Model	Difference
Agam	20.37	63.77	62.14	1.63
Dharmasraya	0.00	4.12	3.92	0.20
Kota Bukittinggi	0.00	63.53	64.18	− 0.66
Kota Padang	71.24	79.79	79.79	0.00
Kota Padang Panjang	4.87	72.96	70.84	2.11
Kota Pariaman	71.29	79.61	81.62	− 2.02
Kota Payakumbuh	0.00	43.71	45.15	− 1.44
Kota Sawah Lunto	0.00	51.38	48.57	2.80
Kota Solok	0.09	60.88	60.08	0.80
Lima Puluh Kota	0.00	36.20	33.21	3.00
Padang Pariaman	95.72	80.77	81.29	− 0.53
Pasaman	2.12	27.67	24.04	3.63
Pasaman Barat	12.10	53.33	51.57	1.76
Pesisir Selatan	9.97	43.85	41.88	1.97
Sijunjung	0.00	25.96	23.89	2.08
Solok	0.93	47.77	44.15	3.62
Solok Selatan	0.00	26.70	23.13	3.57
Tanah Datar	0.30	58.67	55.05	3.62

selection. The Vs30 would likely become a more important factor to consider in finer scale assessments.

6.2 Yogyakarta damage results

To understand the fragility of the housing stock in the Yogyakarta and Central Java region, we need to determine which damage simulations give the most accurate results when compared to the observed damage data. We have identified that, for this case study, the results of the damage simulations are most sensitive to the selection of fragility functions. As such we focus predominantly on how the results vary with fragility function selection and the performance of the fragility functions in this section.

For each of the eleven kabupaten impacted by the 2006 Yogyakarta earthquake, the results for each simulation were compared to the observed proportion of houses damaged in each damage state (Fig. 8). In most of the kabupaten we find that the three fragility function sets listed in Table 5 result in a similar level of agreement with the damage observations, across all damage states. There are some locations and building types where the fit to the data is poor, with either significant overestimations or underestimations of damage in each damage state being simulated. This led us to the develop a Mixed fragility function set. From the damage data we know that wood and bamboo structures performed much better than masonry structures, and initial testing showed that the Martins and Silva (2020) wood fragility functions gave a much better fit with field observations than the HAZUS wood fragility functions. The Martins and Silva (2020) also provide a better fit with field

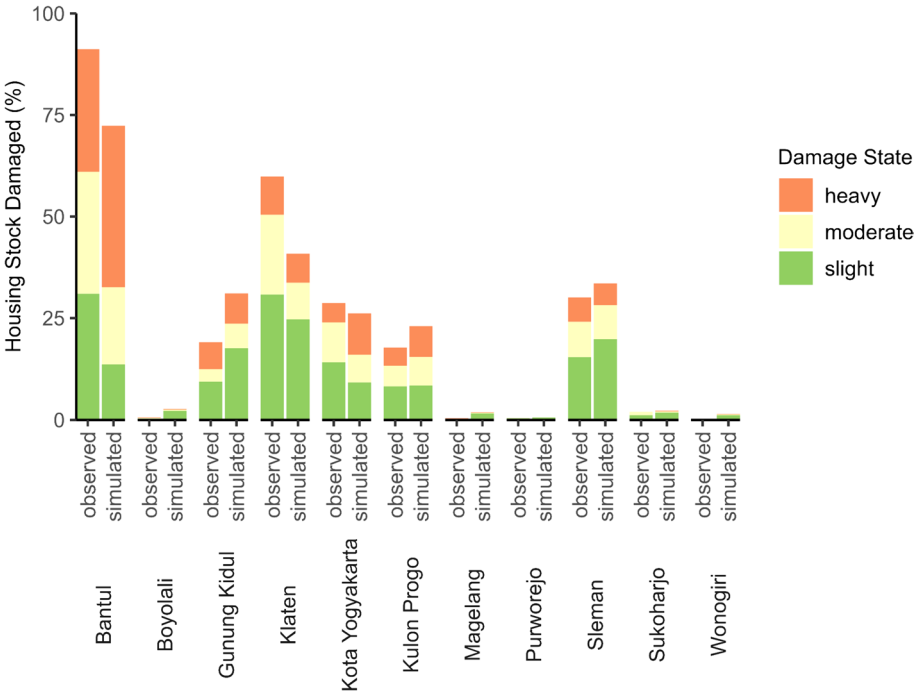


Fig. 8 The observed and the simulated damage by damage state for each of the kabupaten impacted by the 2006 Yogyakarta earthquake. These are the results of 1000 ground motion simulations with the Zhao et al (2016a) GMM and the Cipta et al. (2017) Vs30 model. There is reasonable agreement between the observations and the simulations in each location. These results are from modelling using the best performing fragility functions in each location which differ from kabupaten to kabupaten. The kabupaten with the most damage, Bantul and Klaten, have the poorest result, with the simulations underestimating the damage that was recorded

observations for masonry houses, except when there are high proportions of unreinforced masonry present. However, from the kabupaten of Bantul where we know 95% of the houses are unreinforced masonry, we find that the unreinforced masonry HAZUS function gives better results than all other unreinforced masonry fragility functions considered. Thus, the Mixed fragility functions use the HAZUS functions for unreinforced masonry and the other housing types use the Martins and Silva (2020) fragility functions. Additional simulations were undertaken with these new fragility function assignments and the four selected GMMs.

In general, the fragility functions that agree best with the observed damage data vary from kabupaten to kabupaten. There is an observable pattern in this variation, wealthier and less densely populated locations have damage observations that are consistent with more resilient fragility functions (Martins and Silva 2020; Villar-Vega et al. 2017), the middle class and densely populated regions of Sleman and Yogyakarta have damage observations that are consistent with either the Mixed fragility function, and the poorest and densely populated region of Bantul which is almost entirely unreinforced masonry is consistent with the more fragile HAZUS MUR function. Turning to GMM, the selection is more important proximal to the source (Bantul, Klaten and Gunung Kidul) and is less important more distal to the source. This is likely explained by the larger differences in the maximum

ground motion intensity modelled by the GMM closer to the source. The scenario damage simulation results for the best performing fragility functions for each kabupaten, modelled with the Zhao et al. (2016a) active shallow crust GMM are shown in Fig. 8.

There are some locations where the selection of fragility functions is more complicated, and this may reflect greater variation in the quality of materials and construction practices. Or it could be that the proportions of each building type in the exposure model are not an accurate representation of the buildings present. In Kota Yogyakarta and Kulon Progo, the simulations with the Mixed fragility functions give the best agreement with the observed damage data, but slightly over-estimate the damage. Similarly, in Klaten, the simulations with the the HAZUS functions give the best agreement with the observed damage data, but slightly over-estimate the damage. In Klaten, the simulations using the Mixed fragility functions significantly underestimate the damage, indicating that there are other house types which are more fragile than is described in the Martins and Silva (2020 fragility functions. In Bantul it should be noted that the Chiou and Youngs (2014) with the Japan parameters or the Zhao et al. (2016a) active shallow crust GMM are required to simulate the level of damage observed. Yet in adjacent Gunung Kidul the Cauzzi et al. (2014) GMM, which simulates the lowest ground motion intensities, is required to provide a more acceptable agreement with the damage data. This may be attributed to the difference in the geology of the two locations, with Bantul located at the base of Mt Merapi on soft sediments whereas Gunung Kidul is on a limestone plain (Pawirodikromo 2020).

6.3 Padang damage results

Eighteen kabupaten were included in the damage simulations for Padang, including some kabupaten where there was no recorded damage. The kabupaten included in this analysis can be divided into three categories based on geologic setting; kabupaten along the coastline, kabupaten that are inland and have active volcanoes in or near them, and kabupaten which are further inland and lie beyond the active volcanoes. This distinction is important to explain the damage observations. The coastal kabupaten have the most recorded damage, the kabupaten that are located near active volcanoes have recorded damage, but it is lower than would be expected based on distance from the source, and the kabupaten to the east of the volcanoes have no recorded damage. As mentioned above, only the Abrahamson et al. (2016) intra-slab GMM with a fore-arc and back-arc region defined was used in this study as it is the only one that could predict ground motion that was reflective of the observed damage pattern.

The damage simulations perform best in the coastal kabupaten. In Kota Pariaman where the highest percentage of houses were damaged the simulation with the HAZUS fragility function best fits the observed damage data. In Kota Padang which had the largest number of houses damaged the simulation with the combined fragility functions best fit the damage data. Note that for West Sumatra the combined fragility functions include the HAZUS function for reinforced concrete with brick infill houses as well as unreinforced masonry houses due to the observed poor performance of this building type (Sengara et al. 2010). These results agree with the post disaster surveys which all commented on the particularly vulnerable masonry houses in both cities (Chian et al. 2009; Sengara et al. 2010). Notably in Padang Pariaman landslides destroyed three villages. If we assume the housing stock here is the same as that of Kota Pariaman, the capital of Padang Pariaman, we can surmise that the overestimation of houses in the heavy damage category is those that were destroyed due to landslides. There is less variability in the best fit fragility functions here

compared with the Yogyakarta and Central Java Provinces, indicating perhaps a greater consistency in building style and quality. The scenario damage simulations that agree best with the damage data from each coastal kabupaten are shown in Fig. 9.

The damage simulations did not perform as well in the kabupaten that are more inland from the coast and have active volcanoes nearby. In these kabupaten there was minimal damage recorded. Generally, the simulations overestimate the amount of damage in all damage states by a significant amount. It is unlikely that the houses in these locations are so substantially different in either type of house or fragility to what has been observed in the coastal kabupaten, or even in Yogyakarta and Central Java Provinces. It is more likely that the issue lies with the maximum ground motion intensity that is being simulated for these reasons. It should be noted that the Abrahamson et al. (2016) GMM predicts the lowest ground motion intensity of any of the GMMs initially modelled (see Fig. 5). The kabupaten that lie to the east of the active volcano chain have been defined as lying in the back-arc region in the GMM. There was no damage observed in these locations, but all damage simulations predict damage to occur in these kabupaten. This confirms that the maximum ground motion intensity simulated by the GMM is too high. This indicates that the Abrahamson et al. (2016) GMM with the fore-arc and back-arc region defined, whilst much better than the other GMM tested, does not fully account for the effect of the presence of active volcanism and the increase in attenuation associated with it. It is reasonable to expect a low Q region around volcanoes, even on the fore-arc side, which would increase the seismic attenuation. Perhaps this could be addressed using the Rvolc parameter, which is the seismic wave path distance across a volcanic zone that is included in the Zhao et al. (2016b) GMM, who suggested it is a more sophisticated way to account for active volcanism than the binary fore-arc and back-arc assignment.

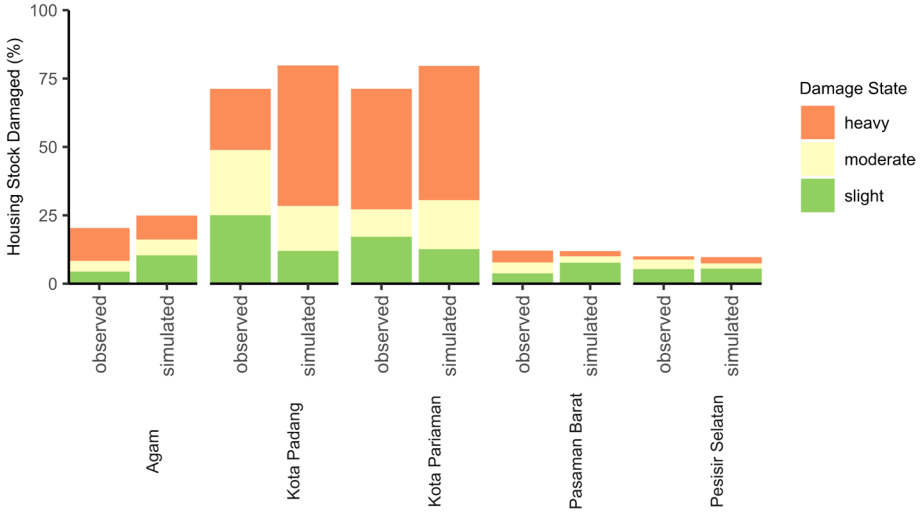


Fig. 9 The observed and simulated damage, by damage state, for the coastal kabupaten impacted by the 2009 Padang earthquake. These are the results of 1000 ground motion simulations with the Abrahamson et al. (2016) GMM and the Cipta et al. (2017) Vs30 model. There is reasonable agreement between the observed and simulated damage in all kabupaten. These are the results with the best performing fragility functions and thus the fragility function used differs from kabupaten to kabupaten

6.4 Comparison of results in most severely impacted kabupaten

The results for the most severely impacted kabupaten in both earthquakes were examined to determine whether there were any similarities in these locations other than proximity to the source. In the 2006 Yogyakarta earthquake the most severely impacted kabupaten were Bantul (DIY) and Klaten (Central Java). In the Padang 2009 earthquake the most severely impacted kabupaten were Kota Padang and Kota Pariaman. Table 11 and Fig. 10 show the indicators used for each of these kabupaten and the exposure model housing profile for each. All kabupaten had a very high proportion of masonry buildings, the key differences are in the poverty decile value and the population density of the kabupaten. Bantul and Klaten were much more densely populated and fall in the lower half of the poverty decile ranking, whereas Kota Padang and Kota Pariaman ranked as some of the wealthiest kabupaten in Indonesia at the time the event occurred. All kabupaten other than Kota Padang also had a high proportion of houses that predate 1980. It should be noted that the housing profile of Bantul was adjusted due to additional field information that all the masonry houses there are unreinforced masonry, which could not have been predicted with the mapping scheme.

The simulated ground motion in all four kabupaten is found to be quite similar (see Figs. 4 and 5). As established, the HAZUS functions give the best results for unreinforced masonry houses across Yogyakarta, Central Java, and West Sumatra. In Klaten other types of masonry house (i.e. confined masonry with low ductility) use the Martins and Silva (2020) fragility functions. As discussed earlier the wealthier kabupaten in Yogyakarta were better represented by the Mixed fragility functions where the Martins and Silva (2020) functions are masonry houses other than unreinforced masonry. However, in Kota Padang and Kota Pariaman the HAZUS functions are applied to all masonry house types to best reproduce the observed damage in these locations. The HAZUS functions are more fragile than the Martins and Silva (2020) for all masonry house types. This shows that population density, relative wealth and setting (urban, rural or city) can be used as indicators to predict the types of houses present, but the general trend of the wealthier a location the more resilient a house of the same type (i.e. better represented by the Martins and Silva 2020 functions rather than the HAZUS functions) does not hold true for Kota Padang and Kota Pariaman.

Table 11 Indicators used in the mapping scheme for the four most severely impacted kabupaten

Kabupaten	Event	Poverty Decile* (10 wealthiest, 1 poorest)	Population density (people/km^2)	Setting (urban/rural/city)	Wall material (%) (BPS 2005, 2009)		
					Masonry	Wood	Bamboo
Bantul	2006 Yogyakarta	5	1620	Urban	95	2.5	2.5
Klaten	2006 Yogyakarta	3	1736	Urban	93	5	2
Kota Padang	2009 Padang	9	1206	City	85	14	1
Kota Pariaman	2009 Padang	9	961	City	84	15	1

*A ranking across Indonesia of the percentage of the population which live below the poverty line. The values used were reported in the post disaster damage reports (BAPPENAS 2006; BNPB, 2009)

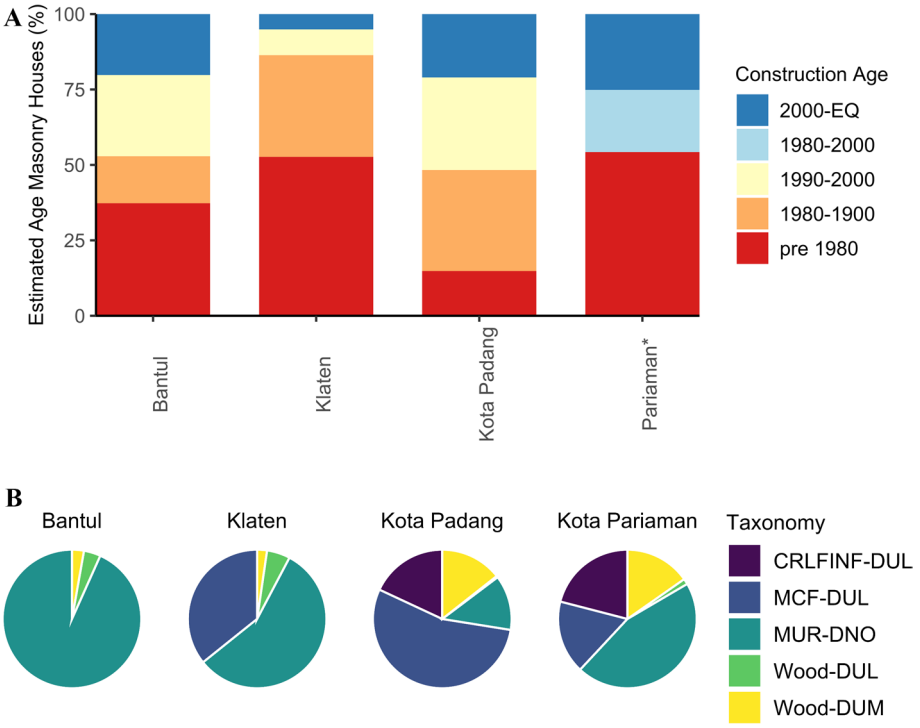


Fig. 10 PanelA shows the estimated construction age of the houses in Bantul, Klaten, Kota Padang and Kota Pariaman. They range from pre 1980 to just before the respective earthquake events. *Pariaman includes data from Kota Pariaman, Padang Pariaman and Mentawai Islands and cannot be disaggregated. Panel B shows the resulting taxonomy and relative proportions of each that was achieved using the mapping scheme with the data from Table 11 and the age profiles

7 Discussion and conclusions

7.1 Sensitivity to input model selection

The results on the sensitivity of input model selection indicate that for both events in this study the damage results are very sensitive to the fragility model selected. This is because there is such large variation between the fragility functions that are for the same broad building type. For example, the range in damage simulation results for Bantul, a location with quite a uniform housing profile, vary greatly depending on the fragility functions used, with all but the HAZUS fragility functions significantly under predicting the level of damage that was observed. Conversely if we look at the simulation results using HAZUS fragility functions for another location, Kota Yogyakarta, we can see that the damage simulated significantly overestimates what was observed (Fig. 11). This highlights the challenges of choosing the fragility functions based only on the type/description of a building, without testing against empirical data. It also suggests that the fragility of unreinforced masonry houses, for example, may very even within a small area such as DI Yogyakarta. This may be due to the quality of materials and construction practices. Or it could be due to variations in geology or soil conditions which are more localized than can be accounted

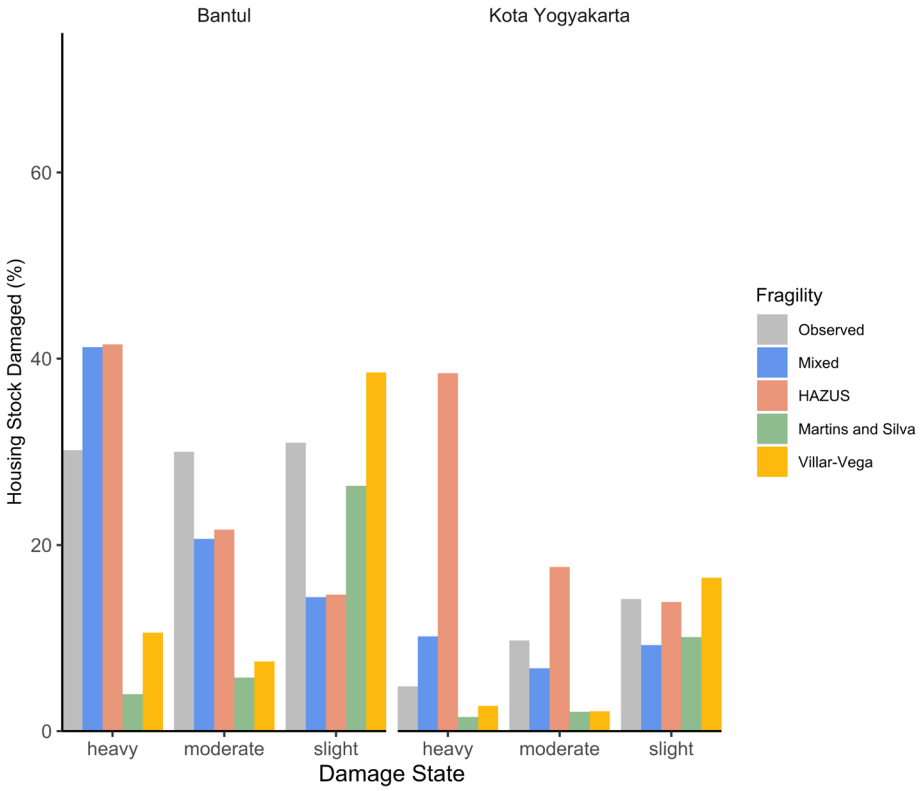


Fig. 11 The sensitivity of the simulated damage results to fragility function selection in Bantul and Kota Yogyakarta. These are the results of 1000 ground motion simulations with the Chiou and Youngs (2014) – Japan GMM. The observed damage is in the hatching and each of the fragility functions used in this study are shown in the coloured columns. The choice of fragility function can cause over or underestimation in each of the damage states

for within the Vs30 model. It can be concluded that what is applicable in one location in Indonesia, may not be relevant elsewhere in Indonesia.

To a lesser degree the results are also sensitive to the selection of the GMM. In Indonesia this is particularly challenging as there are no GMMs for Indonesia due to insufficient ground motion recordings to verify the use of existing GMMs, let alone develop an Indonesia-specific one. For the active shallow crust event in Yogyakarta there are variations in the results based on the GMM selected, and it is observed that this is more important proximal to the epicenter and less important in the more distal locations. This is expected as the greatest variation between the GMM tested in this study is near the epicenter. Variations in performance are likely driven by the specific geology in each kabupaten. Overall, it can be concluded that the simulations using the Chiou and Youngs (2014) GMM with parameters adjusted for Japan gives the best results, followed by the Zhao et al. (2016a) active shallow crust GMM. In Padang however it is the ground motion in distal locations which prove problematic. We conclude that the volcanic arc setting of Sumatra causes increased anelastic attenuation of the seismic waves, thus significantly decreasing the ground motion intensity as the seismic waves move through the active volcanic zones. The only GMM

that can replicate this effect is the Abrahamson et al. (2016) intra-slab function with fore-arc and back-arc regions defined. As discussed above, the Zhao et al. (2016b) GMM has a similar concept with seismic wave path through a volcanic zone, however this is not currently implemented in OpenQuake. Thus, for the Padang event the damage results, particularly distal to the source, are extremely sensitive to the GMM selection. The results suggest that future hazard and risk assessment in Indonesia would benefit from updating the GMM logic-tree that was used in the 2017 PSHA (Irsyam et al. 2020).

The sensitivity to the Vs30 model selection was also considered. It should be noted that the variation between models is small with the Cipta et al. (2017) model generally classifying sites as slightly softer than the USGS model. The results of all the damage simulations only differ marginally depending on which Vs30 model was selected, which is demonstrated in Table 10. Overall, the Cipta et al. (2017) model does contribute to better simulation performance, with the overall softer site classes a better representation of the West Sumatra and Yogyakarta regions considered in this study.

7.2 Predicting the fragility of houses

Given the sensitivity of estimated damage to fragility function selection, it is important to determine whether we can predict this variability in the fragility of houses. The literature and post disaster damage surveys across time and regions in Indonesia indicate that there is little variation in the types of houses that existed at the time of the 2006 Yogyakarta and 2009 Padang earthquakes. The houses can be broadly classified as non-engineered one to two story and either bamboo and/or wood construction with light weight wood cladding, or masonry construction with or without reinforcement (BAPPENAS 2006; BNBP 2009; Boen 2006; Chian et al. 2009). However, it is known that the construction quality and materials can be variable (Boen 2006; Rosyidi et al. 2008). Variation in the fragility of houses of the same broad building type were observed in the 2006 Yogyakarta earthquake and the variation in the damage levels observed near the source has been attributed to the ground motion intensity and the quality of the materials and construction practices (Pawirodikromo 2022). Similarly the damage simulation results suggest that there is variation in the proportion of the house types present and variation in the fragility of unreinforced masonry houses in DI Yogyakarta and Central Java. The results suggest that it may be possible to predict this variation in housing stock and fragility using relative wealth as an indicator.

The exposure model was constructed for both Yogyakarta and Padang using multiple eras of census data combined with literature data post-event and construction practices in Indonesia. The relative wealth, setting (city/urban/rural) and the population density of a kabupaten was used to determine the proportions of different types of masonry houses that exist in a location, including unreinforced masonry, confined masonry, and reinforced concrete with unreinforced masonry infill. In the absence of detailed field exposure data this provides a good approximation of the houses that were present prior to the earthquake events. Preliminary analyses showed this method produced an exposure model that was consistent with the field observations of building types and building performance. Several GMMs, two Vs30 models and multiple sets of fragility functions were considered and tested to determine the sensitivity of the damage simulations to each model selection. In the two case studies the damage simulation results were found to be most sensitive to fragility function selection. Thus, it is critically important to understand how to best select fragility functions.

In the Yogyakarta and Central Java region it was found that the fragility of houses varied. The housing stock in wealthier areas consisted of higher proportions of the more resilient confined masonry houses and the housing stock in poorer areas consisted of higher proportions of unreinforced masonry houses. Specifically in relation to unreinforced masonry houses the fragility was found to vary spatially, with these houses being less resilient in poorer areas than in wealthier areas. This is supported by the damage data which shows that the poorest kabupaten of Bantul and Klaten experienced the most damage, even though other locations were closer to the epicenter. The simulations confirm the assumptions of the exposure model; that the fragility decreases with increasing wealth, that the fragility increases with the age of the building stock, and the fragility increases with increasing population density. Thus it may be possible to use these indicators of relative wealth, construction age, and population density, to estimate the types of houses present and guide fragility function selection for unreinforced masonry houses. These conclusions are in agreement with the statements from BNPB (2009) and BAPPENAS (2006) that poorer people are less likely to be able to afford houses that are seismically resilient.

In West Sumatra we observe that the wealthier areas with moderate density had some of the most fragile building stock. At the time of the 2009 earthquake Kota Pariaman and Kota Padang were ranked among the wealthiest kabupaten in Indonesia, however many post disaster surveys noted the particularly poor quality of construction. The types of houses present in these locations in the exposure model agreed well with the available field data from Chian et al. (2009) and Sengara et al. (2010). However, the fragility of the houses present does not agree with the trends that were observed in Yogyakarta and Central Java. The most impacted regions in both events experienced similar levels of ground shaking and had similar site conditions according to the Vs30 model. The poor performance of the houses in the wealthier Kota Padang and Kota Pariaman is likely due to a combination of local variations in the construction practices and materials and the age of the buildings present. Thus, we can conclude that there are differences between the fragility of the same type of house observed in Yogyakarta and Central Java Provinces in Java, to those in West Sumatra. These differences and the inability to use common indicators to predict this variation shows that there is significant and unknown uncertainty in the use of fragility functions in Indonesia without post event data that can be used for verification.

To determine whether the trends observed in Yogyakarta and Central Java Provinces are applicable across Java would require further studies. There is less damage data available for other earthquake events in Java to support further work. Therefore, to be able to account for the fragility of houses in any future damage or risk analysis for Indonesia a different approach needs to be used.

7.3 Implications for damage and risk assessment

The results also show that fragility function selection can have an extreme impact on estimates of the quantity of houses damaged and the damage state of houses. This agrees with the findings of Riga et al. (2017) that total damage and damage states are particularly sensitive to the fragility functions used in an analysis. Without the ability to use indicators to determine the variation in fragility of houses in a systematic way, another approach is required to account for the uncertainty in assigning a particular building type a fragility function. The conservative approach would be to use the most fragile functions for each house type in an exposure model, which would result in worst case damage and risk analyses. However, these results would be unrealistic and not appropriate for informing disaster

management and policy development. Another approach could be to re-derive the analytical function in the Martins and Silva (2020) database; adjusting the parameters to be more representative of Indonesian houses and including Indonesian ground motion records as is done by Hoyos and Hernández (2021). However, the data required for such undertakings is not available in Indonesia. A final option, which has been suggested as a potential option by Crowley et al. (2020) is to implement a fragility function logic tree to account for the uncertainty in this model selection, as is done with ground motion model selection in PSHA.

What is certain is that damage simulations are extremely sensitive to the selection of fragility functions and that multiple fragility functions are required to represent the same type of house in Indonesia. The fragility of any one type of masonry house can vary from location to location. This variation appears to be correlated with the indicators of poverty, population density and setting in Yogyakarta and Central Java Provinces. These indicators do not work well in West Sumatra, suggesting that there is a significant difference in the construction materials and quality compared to Java, at least in DIY and central Java. This study has shown that it is not possible to use existing models for fragility or ground motion, even when they are for the same type of house or tectonic setting, without robust analysis of the applicability to a specific location and understanding the sensitivity of the results to the decision and the uncertainty in the decision. Further work needs to be undertaken to carry through the findings of this study to risk assessment in Indonesia.

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Data availability The Census data used in this study is freely available to download from the IPUMS, accesed at https://international.ipums.org/international/. The Household Survey data (Susenas) used in this study is confidential and held by the Australian Data Archive on behalf of Badan Pusat Statistik. This data can be requested from BPS https://www.bps.go.id/. The WorldPop data used in this study is freely available to download https://hub.worldpop.org/doi/https://doi.org/10.5258/SOTON/WP00114.

Declarations

Conflict of interest The authors have not disclosed any competing interests.

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